

Modelling Gravitational Lensing and Einstein Rings, Analysis of SLOAN Lens Survey Data set

Project Team Code Y3RF: Cyrus Tang, Joseph Dyson, Leiyi Sun and Prabhmeet Johal

Supervisor: Elias Van den Driessche – PhD student

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Abstract

Using python our team developed a ray tracing model of gravitational lensing to measure the theoretical value of the radius of an einstein ring around a lens and compared the theoretical prediction to those recorded in the Sloan lens ACS survey (SLACS) [1]. We analysed the data to investigate any discrepancies between theoretical and observational data and possible reasons that could explain these differences.

I. BACKGROUND

The main concept of the project was to study gravitational lensing around stellar objects such as stars and produce a computational model of light bending in the presence of a gravitational field using ray tracing technique. This idea was then extended to model einstein rings around a lens and determine the theoretical value of the ring radius for different masses of the lens. Our supervisor advised us to first focus on researching gravitational lensing and identifying the relevant equations and computational methods before continuing. The original idea came from Cyrus. Our group initially determined that there would be 3 prongs to our project:

Initial goals:

- 1) to recreate Einstein's 1919 light deflection experiment computationally, which predicted that a single light beam grazing the sun would be deflected by approximately 1.75 arcseconds.
- 2) to extend the code to a ray tracing model which shoots multiple light beams from the earth to a lens and source.
- 3) to extract the Einstein ring radius from the computational model and repeat for multiple lenses for a certain configuration of the earth, lens and source and compare to the observed value of the ring radius determined from the SLACS data set [1].

Extended goals:

- 4) to make the ray tracing code more efficient by implementing a binary search method to determine which light beams contribute to the Einstein ring.
- 5) to research Euclidean space and non-Euclidean space, investigating possible reasons for the discrepancy between theoretical and observational data.
- 6) to use the model and the existing data to provide evidence for the existence of dark matter.

II. DESCRIPTION OF PROJECT WORK

A. Project start-up

The project commenced on 5th May 2026, with an initial supervisor meeting on 12th May and a group meeting with all four team members a week prior. We continued to have regular meetings with our supervisor, on the 26th May, 2nd June and 5th June as well as regular team meetings to discuss the progress of our project each week. Joseph set up a OneNote lab book so that each member could update their progress on the project and include their contributions towards the project. Prabhmeet set up a shared

OneDrive folder which was used to store the python code and data files so that each team member could work on different aspects of the project and build off each others' code.

B. 1.75 arcsecond recreation and algorithm development

Each of us researched and learned how to utilise RK4 numerical methods in Python [2] and created our own scripts each to model this light deflection experiment with Python, during the week beginning Monday 18th May. It was discovered, that when the star had the same mass and radius as our Sun, the angle of deflection was given as 1.75 arcseconds. This was the famous result of 1919, which was inspired by Einstein's theory of general relativity. Prabhmeet successfully recreated the 1.75 arcsecond angle of deflection using RK4 [2] (a numerical method to solve first order differential equations) in python on 20th May. This was a huge step forwards as we finally had some code working, but in order to implement raytracing, the code needed to be modified so that we could initialise a light ray at any position travelling at any angle and compute the path that it would travel (the previous code could only initialise the light ray travelling tangential to the sun at the distance of closest approach).

C. implementation of ray tracing and analysis of SLACS data set

The following week Joseph and Cyrus worked together on reading the data from the SLACS data set [1] and converted the relevant parameters to useful distances: D_S (distance from earth to source), D_L (distance from earth to lens) and D_{LS} (distance from lens to source), to create a clean set of data which contained the experimental value of the Einstein ring radius for different lens masses. This required extensive research from Joseph and Cyrus as each parameter in the data set needed to be well understood and the correct equations needed to be identified to convert them into these distances. At the same time Leiyi adapted the 1.75 arcsecond code written by Prabhmeet and successfully created a ray tracer program that satisfied the modifications previously outlined.

D. Pursuing extended goals and finalising results

In the last week of May and first few days of June Leiyi was able to code a more efficient algorithm to trace the path of a light ray passing a lens and determine whether it hits the source by using a binary search algorithm, which significantly improved the efficiency of the code and meant that we did not have to ray trace thousands of different light rays at once. When we compared this model to the data, it was far more reliable, and was statistically reliable as a model for gravitational lensing, as it was very close to the true observed data. In this model, we used binary search methods to find the impact parameter for each lens, which meant the run time of the program was far quicker, as we weren't ray tracing for many different angles. This allowed us to extract theoretical values of the Einstein ring radius for multiple different masses of the lens, and Joseph, Cyrus and Leiyi all worked together on the data analysis to produce a graph of Einstein ring radius squared against lens mass, showing both the theoretical model and experimental observations on the same plot to allow for direct comparison. In addition, a residual plot was also produced to analyse potential errors and discrepancies between the data values. For the open day presentation, Joseph worked on the poster, explaining the overall project, where Cyrus is explaining the theoretical aspects of the project such as the relevant equations involved and what they mean, Prabhmeet is explaining the initial implementation of RK4 to computationally recreate Einsteins 1.75 arcsecond prediction and Leiyi is explaining the computational algorithms and approach to solving the problem. Leiyi is also focusing on most of the editing and visuals for the final video presentation. This write up was produced by all four members in equal parts via a shared OneNote/OneDrive document.

E. Application of skills learned in the First Year Lab course

We made use of the following skills that were taught in the First Year Lab course in Terms 1 and 2:

- 1) Many of the coding practising and methods taught in the computing sessions and used throughout labs such as: reading data from files, cleaning data, producing plot of graphs from a data set using matplotlib.
- 2) Analysing data from graphs and data sets to identify potential sources of uncertainty in the data collected and where they stem from.

III. SUMMARY OF RESULTS

All three prongs of the project were successfully implemented, and the code had allowed us to demonstrate how light bends around stellar objects to produce phenomena such as Einstein rings and we were able to use the data from the SLACS data set to compare with our theoretical predictions, and we analysed the differences between them. We found that the experimental data agreed with the theoretical model for values of the Einstein ring radius less than 5 kpc but observed some deviation of the data points from the line of best fit for values greater than 5 kpc. This deviation was explained by the point mass assumption of the model. We also compared stellar and lens mass, finding proof for dark matter.

IV. CONCLUSION

The initial goals of the project were met, as well as the extended goals. While initially, the model was failing us, Cyrus researched online until he found out how we could go about improving our model, and the new model was reliable. The last extended goal was carried out by Cyrus who concluded that by comparing stellar mass to lensing mass, there was evidence for unobservable, dark matter.

V. PURCHASES AND EXISTING ITEMS

A. *Equipment used and purchased*

We took data using Python on our laptops. We accessed the Sloan Lens Survey online, and so had access to this electronically.

We purchased this item:

- General Relativity for the Gifted Amateur by Stephen Blundell, Tom Lancaster, Amazon, £35.48
- Black Holes, the key to understanding the universe by Brian Cox, Jeff Forshaw, Amazon, £17.39

VI. BIBLIOGRAPHY

The literature that we consulted is listed below, with a brief comment on each item's role in the project.

- [1] THE SLOAN LENS ACS SURVEY. V. THE FULL ACS STRONG-LENS SAMPLE, Adam S. Bolton^{2,3}, Scott Burles⁴, L'eon V. E. Koopmans⁵, Tommaso Treu^{6,7}, Rapha"el Gavazzi^{6,8}, Leonidas A. Moustakas⁹, Randall Wayth³, and David J. Schlegel¹⁰
- [2] Fourth Order Runge-Kutta, <https://lpsa.swarthmore.edu/NumInt/NumIntFourth.html>